Introduction to Mechanical Engg. (IME)

Unit 4 & 5

Title: Fundamentals of Thermodynamics **(10 Hrs)**

U1.1. **Fundamental concepts and definitions:** Scope of Thermodynamics, Macroscopic and Microscopic approaches, Systems, Properties, Process, State, Cycle, Thermodynamic equilibrium, Pressure, Energy and its form, Work and heat, Enthalpy.

Laws of thermodynamics:

Zeroth Law of thermodynamics: Concepts of temperature, Zeroth law, Ideal gases and gas mixtures.

First law of thermodynamics: First law of Thermodynamics for closed and open system, Flow processes and control volume, Flow work, Steady flow energy equation.

Second Law of Thermodynamics: Statement of second Law, Heat Engine, Refrigerator, Heat Pump and Carnot cycle, Basic Concept of Entropy.

Application of Thermodynamics: Steam Power Plant, Refrigerators and Heat pump, I.C. Engines (Brief Description of different components of above mentioned systems and working principles with Schematic diagram only).

FUNDAMENTAL CONCEPTS AND DEFINITIONS

1.1 Introduction:

Thermodynamics can be defined as the science of energy. Thermodynamics stems from the Greek words ***therme*** (heat) and ***dunamics*** (motion). There are certain physical processes that occur spontaneously in nature. These are natural processes but there are some other processes which we are forcing to occur. All these processes are not random or zig zag manner. They are in rethym. They are having certain direction and also they obey some laws.

So thermodynamics is defined as a fundamental science that describes the basic laws in relation to different physical processes, which involve transfer or transformation of energy and the relationship among the different physical properties of substance which are affected by such processes.

There are some laws which govern the energy interaction i.e. Zeroth law, first law of thermodynamics, second law of thermodynamics & third law of thermodynamics.

The Zeroth Law deals with thermal equilibrium and provides a means for measuring temperatures.

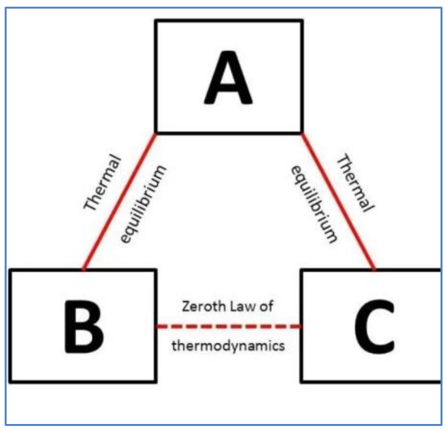


Fig. 1 Zeroth law of thermodynamics

The First Law deals with the conservation of energy and introduces the concept of internal energy.

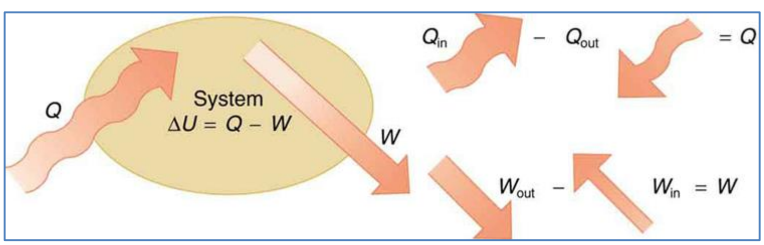


Fig. 2 First law of thermodynamics

The Second Law of thermodynamics provides with the guidelines on the conversion of internal energy of matter into work. It also introduces the concept of entropy.

The Third Law of thermodynamics defines the absolute zero entropy. The entropy of a pure crystalline substance at absolute zero temperature is zero.

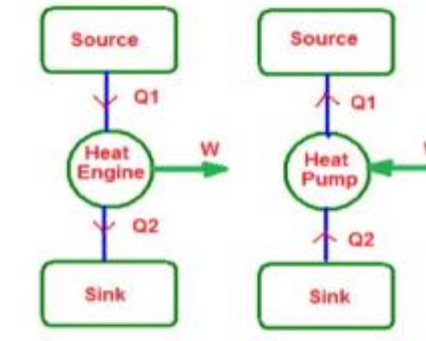


Fig. 3 Second law of thermodynamics

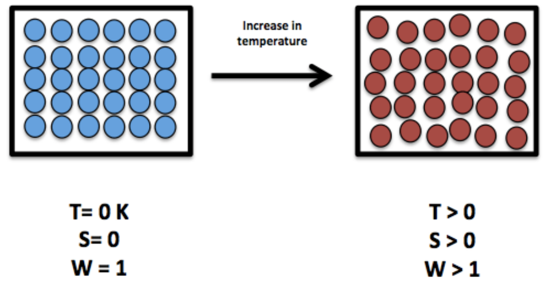


Fig. 4 Third law of thermodynamics

1.2 Application Areas:

- Turbine, compressor, pumps, fans
- Automobile engines
- Refrigeration systems
- Air-conditioning systems
- Fossil & nuclear fuel power plants

1.3 Macroscopic Approach & Microscopic Approach:

There are two views in the study of thermodynamics.

- Macroscopic approach
- Microscopic approach

i. Macroscopic approach: Substance consists of a large no. of particles called molecules. The properties of the substance naturally depend on the behavior of these particles. For example, the pressure of a gas in a container is the result of momentum transfer between the molecules and the walls of the container. However, one doesn't need to know the behavior of the gas particles to determine the pressure in the container. It would be sufficient to attach a pressure gage to the container. This macroscopic approach to the study of thermodynamics that doesn't require knowledge of the behavior of individual particles. It is the overall behavior approach. It is also called as **classical thermodynamics**. It provides direct and easy way to the solution of engineering problems. The macroscopic approach has the following features.

- The structure of the matter is not considered.
- A few variables are used to describe the state of the matter under consideration.
- The values of these variables are measurable following the available techniques of experimental physics. eg: pressure, temperature.

ii. Microscopic approach: A more elaborate approach based on the average behavior of large groups of individual particles is called microscopic approach. This is also called as **statistical thermodynamics**. It is essential for applications involving high speed gas flows, very low temperature etc. The microscopic approach can be summarized as:

- Knowledge of the molecular structure of matter under consideration is essential.
- A large number of variables are needed for a complete specification of the state of the matter.

1.4. System, Surrounding & Boundary:

System: It is a prescribed region or any space or a finite quantity of matter on which we focus our attention to study its properties.

In mechanics, if the motion of body is to be determined normally the first step is to define a free body and identity all the forces exerted on it by other bodies. Newton's second law of motion is then applied. System in thermodynamics is equivalent to free body in mechanics. Once the system is identified and the relevant interactions with other systems are identified, one or more physical laws are applied. So system as a quantity of matter or a region in space chosen for study.

Everything external to be system is called s**urrounding**.

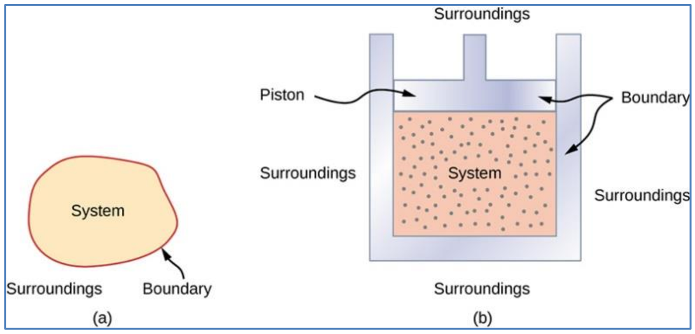


Fig1.5 Thermodynamic system

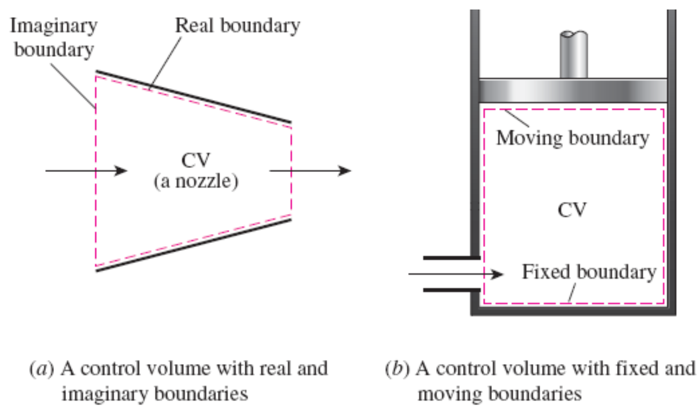


Fig 1.6 A control volume with fixed, moving, real and imaginary boundary

Fig 1.5 shows system, boundary and surrounding. The real or imaginary surface that separates the system from its surrounding is called the boundary which may be at rest or in motion. Fig 1.6 shows a control volume with fixed and moving boundary).

The interactions between a system and its surrounding, which take place across the boundary, play an important part in engineering thermodynamics. Fig 1.6 shows a control volume with real and imaginary boundary)

1.5 Types of system:

Based on the mass transfer and energy transfer across the boundary the systems are divided in 3 types.

- Closed system or control mass system.
- Open system or control volume system.
- Isolated system

1.5.1 Closed system (Control mass system)

A closed system as shown in Fig 1.4 consists of a fixed amount of mass and no mass can cross its boundary i.e. no mass can enter or leave a closed system but energy in the form of heat or work can cross the boundary and the volume of a closed system does not have to be fixed.

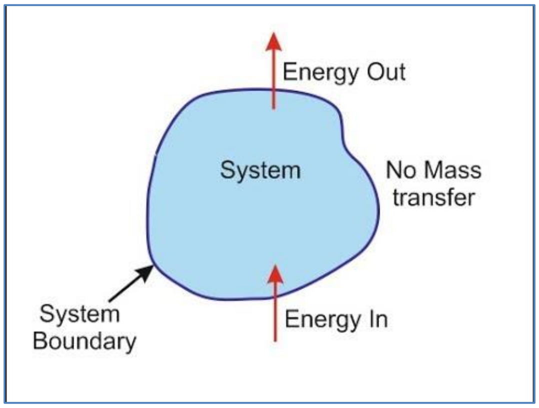


Fig 1.7 A closed system

Example:

Hot coffee in a steel tumbler kept on the table coffee is the system under consideration. It gets cooled after sometimes it rejects its heat to the surrounding. Mass of coffee remains same and there is energy transfer across the boundaries

1.5.2 Open system (Control volume):

An open system as shown in Fig1.5 is a properly selected region in space through which mass can cross the boundary also energy can cross the boundary.

A control volume can be fixed in size and shape or it may involve a moving boundary as shown in figure.

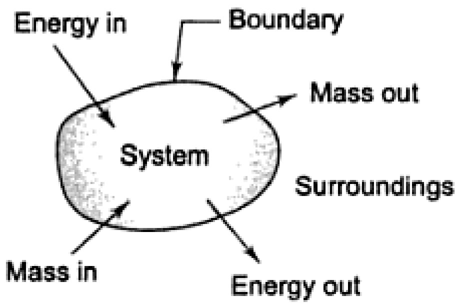


Fig 1.5. An open system

Most control volume however have fixed boundaries and thus don't involve any moving boundaries.

Example: Compressor, turbine, IC engine, nozzle, boilers

1.5.3 Isolated System:

There is neither energy transfer nor mass transfer occurs the boundaries which is shown in Fig 1.6.

Example: Ice cube in a thermos flask, hot gases in a perfectly insulated closed chamber.

System and surrounding together called as universe energy of the universe remains constant.

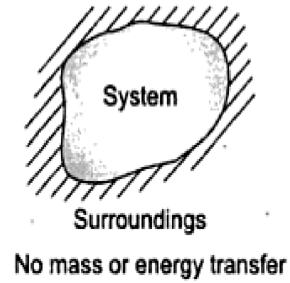


Fig. 1.6. Isolated system

Table 1.1 (Types of system and their mass and energy interactions with surrounding)

Types of system	Mass Transfer	Energy Transfer
Open	✓	✓
Closed	×	✓
Isolated	×	×

1.6. Thermodynamic Properties of system:

A property is a macroscopic characteristic of a system such as mass, volume, energy, pressure, temperature, entropy etc. It is measurable characteristics describing the system. Measureable characteristics are temperature, pressure, Volume etc.

Properties are classified into two categories

(1) Extensive property

(2) Intensive Property

1.6.1. Extensive property: A property is called extensive if its value for an overall system is the sum of its values for the parts into which the system is divided. Extensive properties depend on the size or extent of a system. Example: Mass, volume, energy etc.

1.6.2 Intensive Property:

Intensive properties are not additive in the same previously considered. Their values are independent of the size or extent or mass of a system and may vary from place to place within the system at any moment. Thus intensive property may be function of both position and time where as extensive properties vary at most with time.

Example: Specific volume, specific entropy, temperature, pressure consider an amount of matter that is uniform in temperature & imagine it is composed of several parts. The mass of the whole is the sum of the masses of parts and the overall volume is the sum of the volumes of the parts. However the temperature of the whole is not the sum of the temperature of the parts, it is not same for each part. Mass and volume are extensive but temperature is intensive.

- Extensive Properties per unit mass are called specific properties. Specific properties are designated by small letters. Example: Sp. Volume $\upsilon = \frac{V}{m}$, specific enthalpy h= H/m.
- Ratio of two extensive properties is an intensive property.
- The property of a system depends only on the state of the system and not on the manner in which the state was reached. This means the property of a system should be always an exact differential.
- The no. of properties required to define a system depends on the no. of phases existing in it.

1.7. State:

A set of properties that completely describes the condition of a system is called the state of the system. At a given state, all the properties of a system have fixed values. If the value of even one properties changes, the state will change to a different one. At any instant of time the state is described by its properties such as pressure, temperature, volume etc. Properties when taken on axis to draw a graph they are termed as thermodynamic co-ordinates. At state (1) pressure and volume is (P_1, V_1) and state (2) Pressure & volume is (P_2, V_2) as shown in Fig 1.7

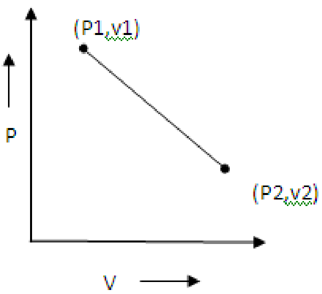


Fig. 1.7.State

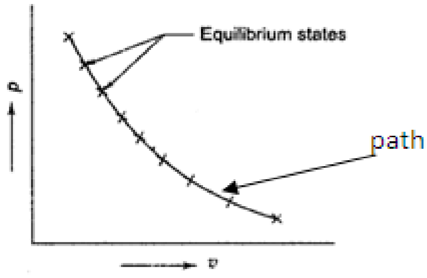


Fig. 1.8

Fig. 1.8. Path

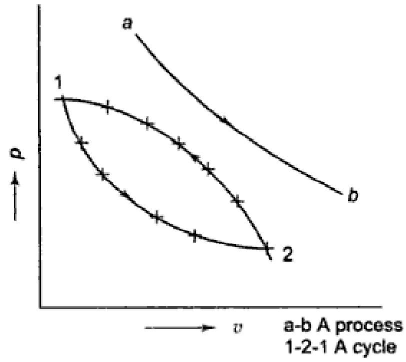


Fig. 1.9.Thermodynamic cycle

1.7.1.Steady state:

If a system exhibits the same values of its properties at two different times i.e. the properties are not changing with time but may be different at different points or positions, then the system is said to be at steady state.

1.8. Path:

When a system undergoes change in its state the line joining the series of intermediate states through which the system has passed is known as path as shown in Fig 1.8.

1.9. Process:

When any of the properties of a system change, the state changes and the system is said to have undergone a process. A process is a transformation from one state to another i.e. a-b as shown in Fig.1.9

1.10. Thermodynamic cycle:

A thermodynamic cycle is a sequence of processes that begins and ends at the same state. At the conclusion of a cycle all properties have the same values they had at the beginning. A cycle is needed for continuous operation and plays an important role in many areas of applications. In Fig.1.9 (1-2-1) is a cycle.

1.11 Quasi-Static Process:

- When a process proceeds in such a manner that the system remains infinitesimally close to an equilibrium state at all times, it is called a quasi-static or quasi-equilibrium process.
- It is a sufficiently slow process that allows the system to adjust itself internally so that properties in one part of the system do not change any faster than those at other parts.

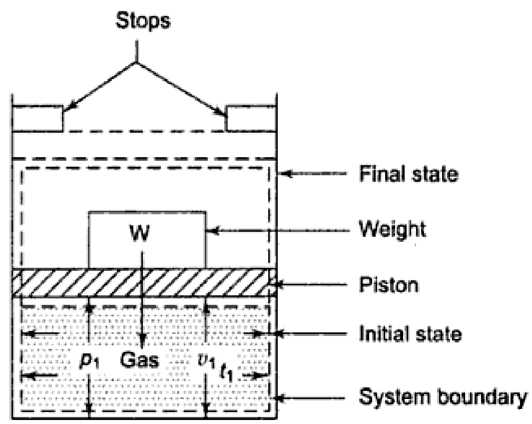


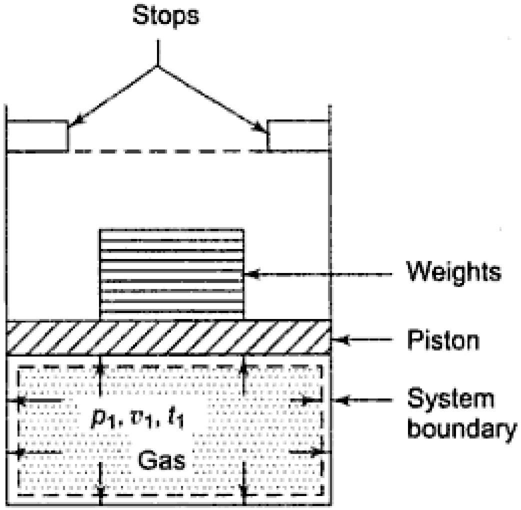
Fig.1.10 (a). Transition between two equilibrium states by an unbalanced force

When a gas in a piston cylinder device is compressed suddenly as shown in Fig.1.10 (a) the molecules near the face of the piston will not have enough time to escape & they will have to pile-up in a small region in front of the piston, thus creating a high pressure region there. Because of this pressure difference, the entire process is non equilibrium or non quasi equilibrium.

- However if the piston is moved slowly as shown in Fig.1.10(b**)** the molecules will have sufficient time to redistribute and there will not be molecule pile up in front of the piston. As a result the pressure inside a cylinder will always be nearly uniform and will rise at the same rate at all locations. So this is a equilibrium process. Here the piston moves in such conditions that the opposing force is always infinitely smaller than the forces exerted on the system.
- On the other hand, if the piston moves under such conditions that the forces are exactly balanced, the system is said to undergo a reversible process. Then the direction of the process can be reversed by an infinitesimal change in pressure across the boundary. A reversible process is always a quasi-static process but the converse is not necessarily true.
- A process is said to be reversible if the system and its surroundings are restored to their respective initial states by reversing the direction of the process. The process is irreversible if it doesn't fulfill this condition.
- Quasi-equilibrium processes are idealized processes and are not a true representation of actual processes. Engineers are interested in quasi-equilibrium processes for two reasons, first, they are easy to analyze and hence at least quantitative information about the behavior of actual systems of interest can often be developed. Second, work-processing devices deliver the most work when they operate on quasi-equilibrium processes and hence serve as standards to which actual processes can be compared.

Also it is instrumental in deducting relationship that exists among the properties of systems in equilibrium.

- In an equilibrium process each equilibrium state is represented by a point on the property diagram and hence the process path can be drawn by a continuous line showing each state.



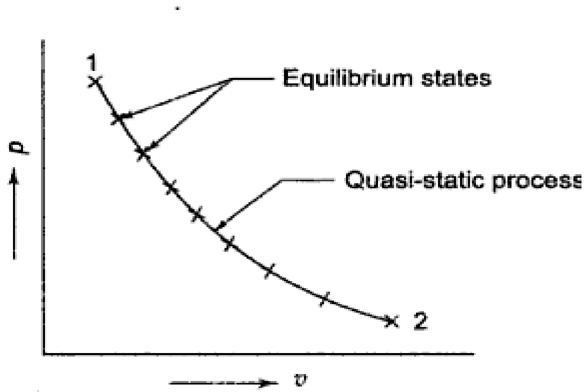


Fig.1.10 (b). Infinitely slow transition of a system   Fig.1.10(c). Quasi equilibrium process

1.12. **Reversible & Irreversible Process:**

Reversible Process:

It is a process in which a system returns to its initial state along the same path without leaving any traces of its being happened on the surrounding. This is possible only when a system passes through a continuous series of equilibrium states. A reversible process is always a quasi-static process.

Irreversible process:

A process is irreversible of a system passes through a series of non-equilibrium states i.e. if the irreversible process is made to reverse the path it will not restore to its initial state. Thus when the process is not reversible we call it as an irreversible process.

Example: (1) Heat transfer between two bodies.

Heat flows from high temperatures body to low temperature on its own. For reverse process to occur i.e. to transfer from low temperature body to high temperature some work must be expanded making this also irreversible. (1) Friction (2) Free expansion (3) Mixing

No process in nature is truly reversible but some may approach reversibility very closely

Example: (1) Frictionless expansion or compression of gas (2) Electrolysis (3) Extension and compression of a spring (4) Isothermal expansion or compression.

1.13. Thermodynamic Equilibrium:

A system is said to exist in a state of thermodynamic equilibrium when no spontaneous change in any macroscopic property is observed as the system is isolated from its surroundings. Equilibrium means the state of balance of a system within itself and between system and surroundings. Thus for attending a state of thermodynamic equilibrium following three types of equilibrium states the system must achieve.

1.13.1 Thermal Equilibrium:

A system is in thermal equilibrium if the temperature is same throughout the system i.e. the system involves no temperature differential, which is the driving force for heat flow.

1.13.2. Mechanical Equilibrium:

It is related to pressure and a system is in mechanical equilibrium if there is no change in pressure at any point of the system with time. However, the pressure may vary with in the system with elevation as a result of gravitational effects. But there is no imbalance of forces. The variation of pressure as a result of gravity in most thermodynamic systems is relatively small and usually disregarded.

1.13.3. Chemical equilibrium:

A system is in chemical equilibrium if no chemical reaction or transfer of matter from one part to another within the system takes place either through diffusion or any other chemical process. Chemical composition of the system doesn't change with time.

Example: Consider a tank containing a mixture of N_2 and H_2 at a sufficiently high pressure and temperature, where the chemical reaction between N_2 and H_2 is feasible leading to the formation of NH_3. If we start with a mixture of pure N_2 & H_2 (in the pressure of catalyst), as time progress we notice the formation of NH_3. The composition of the mixture undergoes a change continuously with time. If the system is isolated and left to itself, after a sufficiently long time, the composition of the mixture reaches a steady state value and there is no further change in the composition of the mixture. The system is said to be chemical equilibrium. A system achieves thermodynamic equilibrium only when it satisfies the above three equilibrium conditions. Thermodynamic analysis deals with the equilibrium states only.

Table 1.2: Criteria for an equilibrium:

Equilibrium	**Criteria**

Thermal	Equality of temperature
Mechanical	Equality of pressure & force
Chemical	Equality of chemical Potential
Thermodynamic	All the above

1.14. Pressure:

A fluid element or mass is essentially acted upon by two categories of forces: body forces and surface forces. Body forces on fluid elements are caused by agencies such as gravitational, electric or magnetic fields. The magnitude of these forces is proportional to the mass of the fluid. Surface forces represent the action of the surrounding fluid on the element under consideration through direct contact. The normal force per unit area is called the intensity of pressure. Pressure is always compressible. Pressure is related to fluids (liquid & gas). A force acting on portion δA of a surface can be resolved into 2 components; i.e. one normal and other tangential to the area δA. The normal force per unit area is called intensity of pressure. From continuum view point, pressure is defined, at a point as:

$$\sigma = \lim_{\delta A \to 0}\left(\frac{\delta F_n}{\delta A}\right)$$

Where, δFn is the force acting normal to the δA.

P= Pressure at a point

A= surface area on which normal force (F_{nor}) is acting.

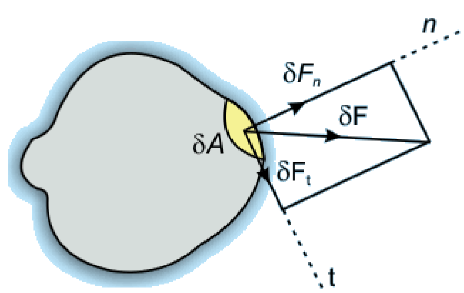


Fig.1.21

1.14.1 Units:

1 pascal (Pa) = 1 N/m²

1 bar = 10^5 N/m² = 10^5 Pa, 1 KPa = 10^3 Pa

1 atm (standard atmospheric pressure) = 1.01325 bar= 101325 Pa=760 mm of Hg = 760Torr

1 MPa= 10^6 Pa

1 Kgf/cm² = 9.81 N/cm² = 9.81 X 10^4 pa = 0.981 bar.

The pressure executed by 1 mm column of mercury is named as Torr.

750 mm of mercury = 1 bar = 100 KPa.

1.14.2. Absolute Pressure, Gauge Pressure & Vacuum Pressure:

Atmospheric Pressure: Atmospheric pressure is pressure exerted by surrounding air on the surfaces of the body with which it is in direct contact. The atmospheric pressure varies with temperature and altitude above sea level.

Gauge Pressure: The difference between the absolute pressure & local atmospheric pressure is called gauge pressure.

Vacuum Pressure: A pressure reading below the atmospheric is known as vacuum, rarefaction or the negative pressure as shown in Fig 1.22.

Absolute Pressure: The actual pressure at a given position measured relative to absolute vacuum. Zero pressure intensity will occur when molecular momentum is zero. Such a situation can occur only when there is a perfect vacuum, i.e. a vanishingly small population of gas molecules or of molecular velocity. $P_{abs}=P_{atm}+P_{gauge}$ & $P_{abs}=P_{atm}-P_{vacum.}$

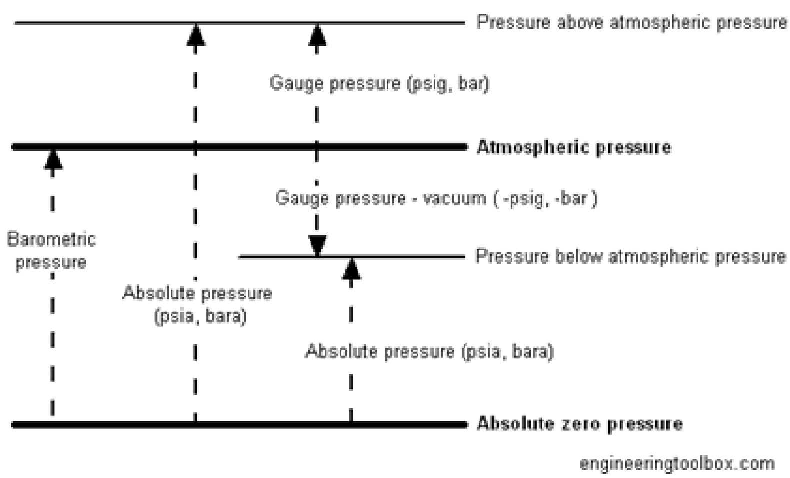


Fig1.22.Relationship between pressure

1.14.3. Hydrostatic Pressure:

- Pressure at any point in a fluid at rest is same in all directions, that is, it has magnitude but not a specific direction and can be treated as a scalar quantity. However, the pressure can vary from point to point. For example- variation of atmospheric pressure with elevation and pressure variation with depth in ocean, lakes and other bodies of water.

- Pressure in a fluid increases with depth because more fluid rest on deeper layer and the effect of this extra weight on deeper layer is balanced by an increase in pressure.

According to hydrostatic law of pressure

$$\frac{dP}{dZ} = -\rho g$$

$$\Rightarrow \int_1^2 dP = \int_1^2 \rho g \; dz$$

$$\Rightarrow (P_2 - P_1) = -\rho g (z_2 - z_1)$$

$$\Rightarrow P_2 - P_1 = \rho g h. (h \text{ is measured from the free surface})$$

$$\Rightarrow P_{abs} = P_o + \rho g h$$

$$where P_0 = atm. \text{Pr}essure.$$

1.15. ENERGY TRANSFER & WORK TRANSFER

1.15.1. Introduction:

Energy: It is the ability to cause any change. Energy interaction brings about changes in the properties of the system. Energy is of two types: (1) stored energy (2) transition energy

Transition energy: Transition energy is that form of energy which can flow across the system boundary in both directions. Work and heat are transition form of energy so they are called energy in transit. Work and heat are transient form of energy which got its identification at the system boundary. Transition energy transfer from system to surrounding or form surrounding to system or from system to system as a result of potential difference. Ex. Temperature gradient difference results in the transfer of heat pressure gradient results in mechanical work.

Stored energy: It is present in the system by virtue of its orientation in a force field, its chemical composition, atomic and molecular structure. Molecules during motion may collide with other molecules and with the walls of the container as a result of which, rotation and vibration of molecules takes place. Molecules may possess transitional energy on account of translational motion, rotational energy on account of their rotational motion, and vibration energy on account of vibration motion. Thus internal energy of the system is the combination of translational, rotational and vibration energy.

1.15.2. Work interaction (Definition & Calculation)

Work is done by a force as it acts upon a body moving in the direction of the force as shown in

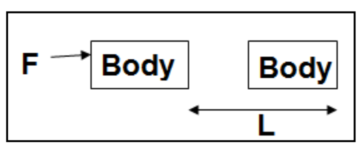
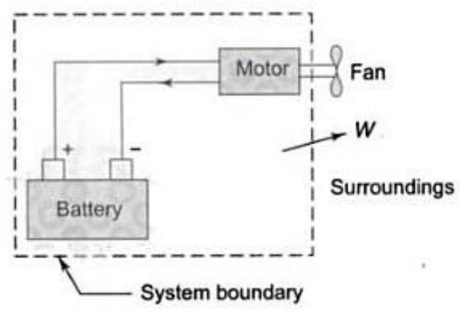


Fig. 1.28 (a)

Fig 1.28 (a), W= F X L

Where, W = work done, F= force applied, L= distance moved (Unit = N- m or Joule)

In thermodynamics work transfer occurs between the system and the surroundings. Work is said to be done by a system if the sole effect on things external to the system can be reduced to the raising of a weight.



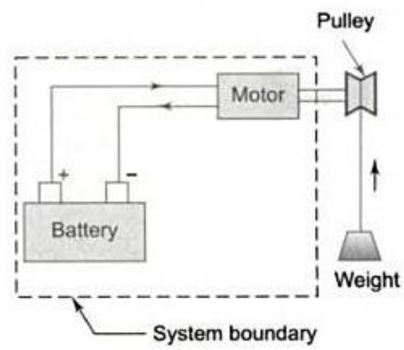


Fig 1.28 (b). Battery- motor system driving a fan Fig 1.28 (c). Work transfer from a system

Consider the battery and motor in system as shown in Fig 1.28 (b) the motor is driving a fan. The system is doing work upon the surrounding. When a fan is replaced by a pulley as shown in Fig 1.28 (c) the weight may be raised with the pulley driven by the motor. The sole effect on things external to the system is then the raising of a weight.

1.15.3. Sign convention:

If the work is done by the system on the surrounding the work is said to be positive i.e. work output from the system is positive. Ex. Expansion of fluid

If the work is done on the system by the surrounding work is said to be negative i.e. work input to the system is negative. Ex. Compression of fluid

Heat transfer into the system is positive and heat transfer out of the system is negative.

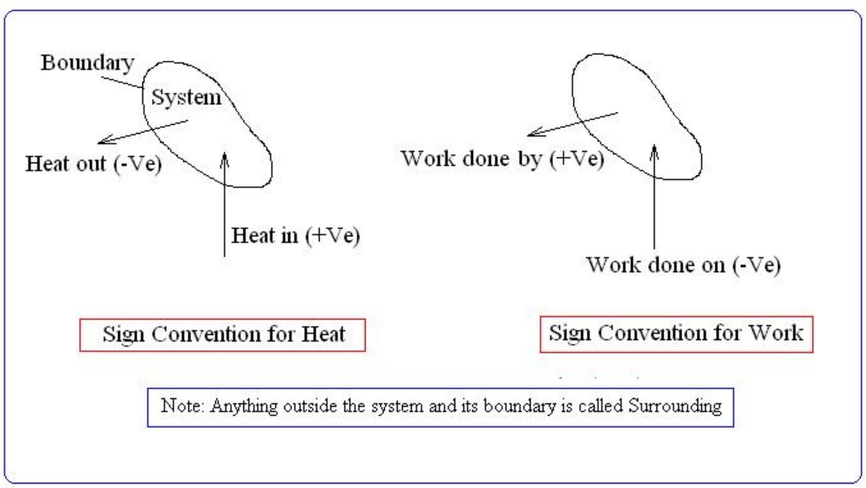


Fig.1.29 Sign Convention

1.15.4. Work is not a property:

$$W = \int_{S1}^{S2} F.ds$$

- To evaluate the above integral, it is necessary to known how the force varies with the displacement. So the value of W depends on the details of the interactions taking place between the system and surroundings during a process and not just the initial and final states. So work is not a property of the system or the surroundings.
- Work is a path function and mathematically expressed as inexact differential

$$\int_1^2 \delta w = W_{1-2} \neq W_2 - W_1$$

Where, δ stands for inexact differential.

W_{1-2} is the work associated with the path as the process passes from state 1 to state 2.

Cyclic integral of a property is always zero.

$$\oint dv = 0, \quad \oint dp = 0$$

1.15.5. Mechanical Work:

1.15.5.1. P.dv work or displacement work (Non-flow work)

- The process of compression and expansion of gas in a piston-cylinder system is one of the mechanical work transfers.
- Let us consider a mass of gas as shown in Fig 1.30 (a) enclosed within, the space in a piston-cylinder system. The system is considered to be a closed system since no gas can leave the system. Let at a particular position of piston say (1)-(1), the initial pressure of the gas is P_1 and volume of the gas is V_1. Let the system is in thermodynamic equilibrium.
- Piston moves outward due to gas pressure inside the cylinder where external resistance is only in smaller than that produced by the pressure within the system.
- Let the piston moves to a new position defined by section (2)-(2) due to expansion of gas having pressure P_2 and volume V_2.
- Let the pressure and volume of the gas at any intermediate. State by the section (0)-(0) be P and V respectively.

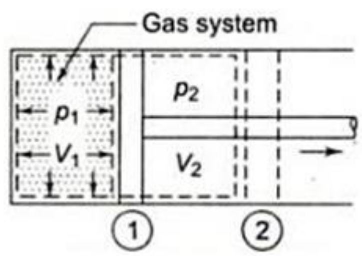


Fig 1.30 (a)

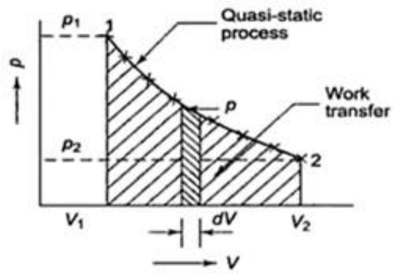


Fig 1.30 (b)

Let A = Cross sectional area of the piston

F= force acting on the piston = P.A

dl=Displacement of piston due to force F

dW= Infinitesimal work done

dW= F.(dl) = P.A. dl = Pdv

Thus the amount of work (w) done by the system due to expansion Process is given by

$$W_{1\text{-}2} = \int_{v1}^{V2} Pdv$$

The integration $\int pdv$ can be performed only on a Quasi-static path condition as shown Fig 1.30.

(b) which the equation, $W1_{\text{-}2} = \int_{v1}^{V2} Pdv$ is valid.

It must be a closed system. There should not be any viscous effect within the system. All the intermediate states the system passes through must be in thermodynamic equipment. There should not be gravity effects or magnetic effects.

1.15.5.2. Pdv work in various Quasi-Static Process:

(a) Constant pressure process:(Isobaric Process):

During constant pressure process (1-2) as shown in Fig 1.31 (a)

work done becomes $W_{1\text{-}2} = \int_{v_1}^{v_2} pdv = p(v_2 - v_1)$

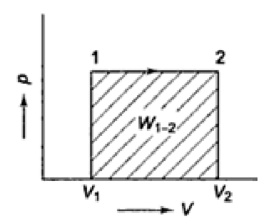


Fig 1.31 (a) Constant pressure process

(b) **Constant volume process (Isochoric process)** :

During constant volume process (1-2) the work done becomes $W_{1\text{-}2} = \int_{v_1}^{v_2} pdv = 0$

Volume constant means dv = 0

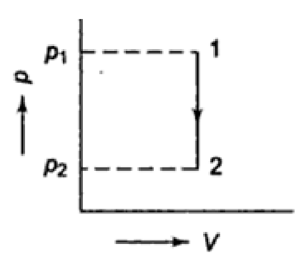


Fig 1.31 (b) constant volume process

(c) Process in which pv =c

In this process 1-2, work done becomes

$W_{1-2} = \int_{v_1}^{v_2} pdv$

$PV = P_1V_1 = P_2V_2 = C$

$P = P_1V_1/V$

Work done becomes $W_{1-2} = \int_{v_1}^{v_2} pdv$

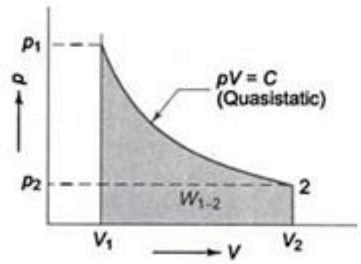


Fig 1.31 (c) process PV=C

$\int_{v_1}^{v_2} \frac{P_1 v_1}{v} dv = p_1 v_1 \ln V_2/V_1$

$= p_1v_1(\ln V_2 - \ln V_1) = p_1v_1 \ln (V_2/V_1)$

but $p_1v_1 = p_2v_2$

$\Rightarrow V_2/V_1 = P_1/P_2$

So work done during the process 1-2 can be written as

$W_{1-2} = p_1v_1 \ln (V_2/V_1)$ or

$= p_1v_1 \ln (P_1/P_2)$

(d)Process in which $pv^n = c$,

Where, n = constant Quasi equilibrium expansion and compression process can represented analytically as pv^n = const. and is called polytropic process.

$pv^n = c = p_1v_1^n = p_2v_2^n$, $p = \frac{p_1 v_1^n}{v^n}$

$$\int \frac{p_1 v_1^n}{v^n} dv$$

Work done during the process 1-2, $W_{1-2} = \int_{v_1}^{v_2} pdv = = p_1 v_1^n \int_{v_1}^{v_2} v^{-n} dv = p_1 v_1^n \frac{v^{-n+1}}{-n+1}$

$$p_1 v_1^{\,n}\left(\frac{v_2^{\,1-n}-v_1^{\,1-n}}{1-n}\right)$$

$$=\frac{p_1 v_1^{\,n}}{1-n}\left(v_2^{\,1-n}-v_1^{\,1-n}\right)$$

$$=\frac{p_1 v_1\times v_2^{\,1-n}-p_1 v_1\times v_1^{\,1-n}}{1-n}\left(\because\left(p_1 v_1^{\,n}\times p_2 v_2^{\,n}\right)\right)$$

$$=\frac{p_1 v_1-p_1 v_1}{1-n}=\frac{p_1 v_1-p_2 v_2}{n-1}$$

$$=\frac{p_1 v_1}{n-1}\left[1-\frac{p_2 v_2}{p_1 v_1}\right]$$

$$W_{1-2}=\frac{p_1 v_1}{n-1}\left[1-\left(\frac{p_2}{p_1}\times\frac{v_2}{v_1}\right)\right]---------(1)$$

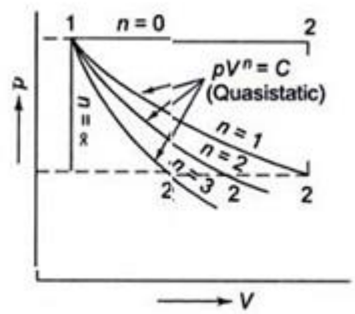


Fig 1.31 (d) (Process PV^n=C)

But $p_1v_1^n = p_2v_2^n$

$$=\left(\frac{v_2}{v_1}\right)^n=\frac{p_1}{p_2}$$

$$\Rightarrow\left(\frac{v_2}{v_1}\right)=\left(\frac{p_1}{p_2}\right)^{\frac{1}{n}}$$

$$W_{1-2}=\frac{p_1 v_1}{n-1}\left[1-\frac{p_2}{p_1}\times\left(\frac{p_1}{p_2}\right)^{1/n}\right]$$

$$=\frac{p_1 v_1}{n-1}\left[1-\frac{p_2}{p_1}\times\frac{p_1^{\,1}n}{p_2^{\,1}n}\right]$$

$$=\frac{p_1 v_1}{n-1}\left[1-\left(\frac{p_2^{\,1-\frac{1}{n}}}{p_1^{\,1-\frac{1}{n}}}\right)\right]$$

$$=\frac{p_1 v_1}{n-1}\left[1-\left(\frac{p_2^{\,\frac{n-1}{n}}}{p_1^{\,\frac{n-1}{n}}}\right)\right]$$

$$=\frac{p_1 v_1}{n-1}\left[1-\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}\right]$$

Putting the value of (v_2/v_1) in the above equation (1)

So work done during polytropic process is $\dfrac{p_1 v_1-p_2 v_2}{n-1}$ Or $\dfrac{p_1 v_1}{n-1}\left[1-\left(\dfrac{p_2}{p_1}\right)^{\frac{n-1}{n}}\right]=\dfrac{R(T_1-T_2)}{n-1}$

During isentropic process pv^K = const.

Where $K = C_p/C_\mu$= (specific heat ratio)

$$W_{1-2} = \frac{p_1 v_1 - p_2 v_2}{k-1} = \frac{R(T_1 - T_2)}{k-1}$$

1.15.6. Heat transfer:

- Heat is the form of energy in transit due to temp. Difference.
- In steady state heat transfer temperature at any location on the system doesn't vary with time. T= f(x,y,z)
- In unsteady state heat transfer, temperature varies with time as well as position. T= f (x,y,z,t)
- Heat transfer into the system is positive and out of the system is negative.

1.16. Modes of Heat transfer:

1. Conduction
2. Convection
3. Radiation

1.16.1. Conduction:

Conduction is the transfer of energy from a more energetic particle of substance to the adjacent less energetic particle as a result of interaction between the particles. Conduction can takes place in solids, liquid & gases. In liquid and gases conduction is due to the collision of molecules during their random motion. In solids it is the combination of the vibration of the molecules in a lattice and the energy transported to the free elections. All good conductors of electricity are good conductor of heat but the reverse is not true. Ex. Diamond. According Fourier's law the rate of heat conduction Qn in a direction n through a layer of thickness Δx is proportional to the temperature difference. At cross the layer and area A normal to the director of heat transfer and is inversely proportional to the thickness of layer.

Along x direction as shown in Fig 1.32 (a)

$Qx \propto \Delta T$

$Q x \propto A$

$Qx \propto 1/\Delta x$

$Qx = -K A (\Delta T/\Delta x)$

Where, Qx = rate of heat transfer along x direction

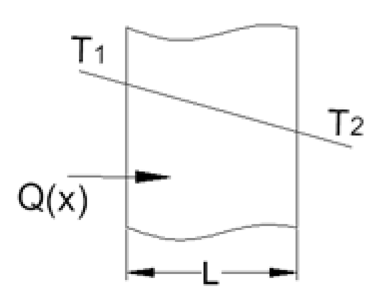


Fig 1.32 (a)

ΔT = temperature difference

Δx = thickness of the layer

A = Area normal to the direction of heat flow

When, Δx =0

$Qx = -KA (\Delta T/\Delta x)$

$Qx = -K A (\Delta T/\Delta x)$

1.16.2. Convection:

It is the mode of heat transfer between a solid surface and the adjacent liquid or gas (fluid) that is in motion. It is involves the combined effect of conduction and fluid motion (advection). Convection is of 2 types (1) forced convection (2) free convection.

Forced convection: convection is called forced convection of the fluid flows is caused by an external means such as fan, pump, blower etc.

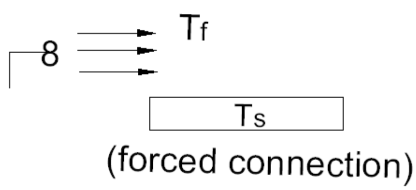


Fig. 1.33

Natural convection: convection is called natural convection or free convection if the fluid motion is caused by buoyancy force included by density difference, due to variation of the temperature of the fluid. Fig. 1.34 shows natural convection.

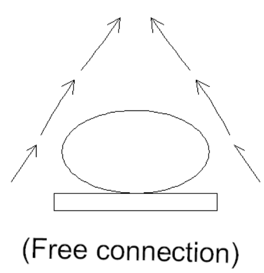


Fig. 1.34

Newton's Law of cooling:

It states that the rate of heat transfer per unit area is directly proportional to the temperature difference between surfaces and fluid. $\frac{Q}{A}\alpha\left(T_S - T_\infty\right)$

Where, Q = rate of heat transfer

T_S = surface temperature °C

T_∞ = fluid temperature °C

A= Area for convection heat transfer (m²)

$$\frac{Q}{A} = \left(T_S - T_\infty\right)$$
$$= Q = hA\left(T_S - T_\infty\right)$$

h= constant proportion ting called heat transfer coefficient (W/m²K)

Q= h.A.($T_S - T_\infty$) = $Q = \frac{T_S - T_\infty}{Y_{ha}} = \frac{T_S - T_\infty}{\left(R_{th}\right)_{conv.}}$

1.16.3. Radiation:

It is the energy emitted by a matter in the form of electromagnetic waves as a result of change in the electronic configuration of atoms or the molecules.

Thermal radiation: The radiation by a body because of its temperature is called thermal radiation.

Black body:

It is an ideal body which absorbs and emits radiation of all wavelengths in all directions it is a good absorber as well as a good emitter. A body colored black is approximately as a black body.

Stefan Boltzmann Law:

It states that the rate of radiation heat transfer per unit area from a black surface is directly proportional to the fourth power of the absolute temperature of the surface.

$$\frac{Q}{A}\alpha\ T_S{}^4 \Rightarrow \frac{Q}{A}\alpha\ \sigma\ T_S$$

Where, T_S= absolute temp. of the surface ink

σ= constant proportionality called Stefan Boltzmann constant (5.67 x 10^{-8} W/m²K^4)

Radiation emitted by a real body at a temp. T_S

Q= σA ϵT_S^4 & Where, ϵ= emissivity of the surface.

1.16.4. Enthalpy:

- Enthalpy is a measure of the total energy of a thermodynamic system.
- It includes the internal energy which is energy required to create a system and the amount of energy required to make room for it by displacing its environment and establishment its volume and pressure.
- $H = U + PV$
- $h = u + pv$ (enthalpy per unit mass is called specific enthalpy)
- It is an intensive property.
- Total enthalpy is $H = mh$.

ZEROTH LAW OF THERMODYNAMICS

1.17. Temperature:

Temperature is defined as the degree of hotness or coldness of a body measured on a definite scale.

- Temperature is the driving force or potential for heat transfer.
- It is a measure of mean K.E. of the molecules of the system. A change in temperature of a system accounts for change in molecular motion and the K.E and the molecule.
- Temperature is a parameter which determines whether or not a system is in thermal equilibrium with another system.
- When work or heat is supplied to a system it is not mandatory that the temperature of the system will increase. It may or may not increase until the molecular K.E. increase.

Example: Temperature of a gas in a container doesn't increase when we put the container in a train.

1.17.1. Zeroth law of Thermodynamics:

If two bodies A and B are separately in thermal equilibrium with third body 'C' then 'A & B' must be in thermal equilibrium with each other as shown in Fig.1.35. Adiabatic wall is a wall which does not allow the heat to pass through it. Let both A & B are separated from C through a diathermic wall.

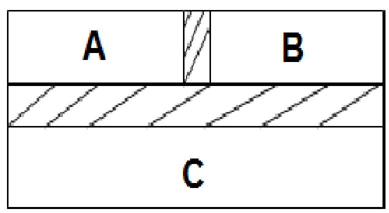


Fig.1.35.Zeroth law of thermodynamics

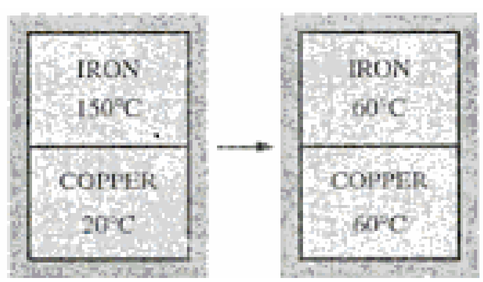


Fig 1.36

Example: A cup of hot coffee left on the table eventually cools off and a cold drink eventually warms up i.e. when a body is brought in contact with another body i.e. at a different temperature heat is transferred from a body of higher temperature to the one at lower temperature until both the bodies attain the same temperature (Fig. 1.36). At that point the heat transfer stops and the two bodies are said to have reached thermal equilibrium. The equality of temperature is the only requirement for thermal equilibrium.

1.17.2. Thermometric Property:

Fortunately, several properties of materials change with temperatures in repeatable & predictable way and this forms the basis for accurate temperature measurement.

Anybody with at least one measureable property that changes as its temperatures changes can be used as a thermometer. Such a property is called a thermometric property and the substance is called thermometric substances.

Measuring Devices	Thermometric Property
1. Liquid-in-glass thermometer.	Length of the liquid in the capillary tube.
2. Const. volume gas thermometer.	Pressure of the gas.
3. Const. pressure gas thermometer.	Volume of the gas.
4. Thermocouple	Thermal emf between two dissimilar metals
5. Resistance thermometer	Resistance

1.18. IDEAL GAS

1.18.1. Introduction:

A hypothetical gas which obeys the law p = RT at all pressure and temperatures is called an ideal gas. The actual volume of the molecules is negligible in comparison to the volume of the container. An ideal gas or perfect gas has no intermolecular force of attraction or repulsion between the gas molecules.

1.18.2. Compressibility factor (Z):

Gases deviate from ideal gas behavior significantly at steps near the saturation region & the critical point. This deviation from ideal gas behaviour at a given temperature and pressure can be accurately be accounted for by the introduction of a correction factor called compressibility factor.

$P v = RT$

$\Rightarrow Pv = ZRT$

$\Rightarrow Z = Pv/RT$

Z= I for ideal gas

$Z<1$, $Z>1$ Real gas

Real gas approaches ideal gas when $P \to 0$and $T \to \infty$

<u>**FIRST LAW OF THERMODYNAMICS**</u>

1.19. First Law of Thermodynamics:

The first law of thermodynamics, also known as the conservation of energy principle, provides a sound basis for studying the relationships among the various forms of energy and energy interactions. Based on experimental observations, the first law of thermodynamics states that energy can be neither created nor destroyed during a process; it can only change forms. Therefore, every bit of energy should be accounted for during a process. A series of Experiments carried out by Joule between 1843 and 1848 from the basis for the **First Law of Thermodynamics.**

The following are the observations during the Paddle Wheel experiment shown in Fig. 1.43.

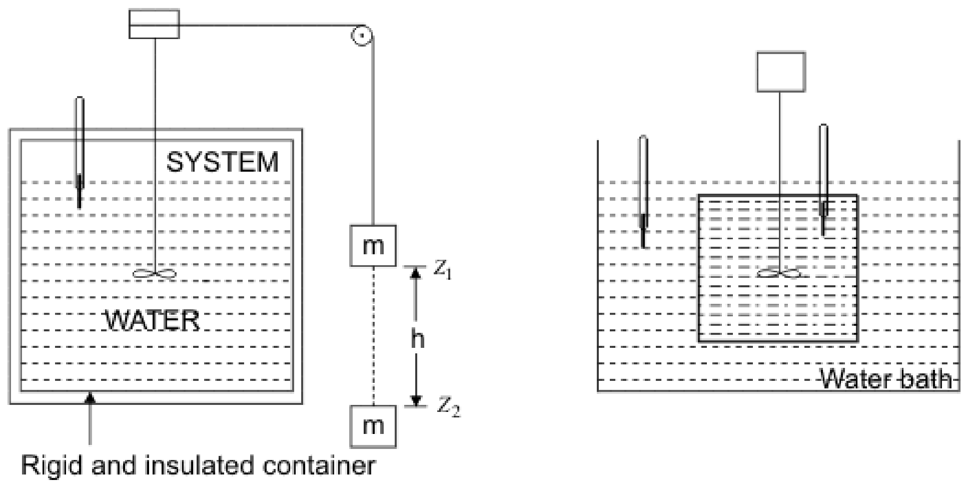


Fig 1.43

- Work done on the system by lowering the mass *m* through (Z_1-Z_2) = change in PE of *m*
- Temperature of the system was found to increase
- System was brought into contact with a water bath
- System was allowed to come back to initial state
- Energy is transferred as heat from the system to the bath

The system thus executes a cycle which consists of work input to the system followed by the transfer of heat from the system.

$$\underset{\text{from the system}}{\oint dQ} = \underset{\text{on the system}}{\oint dW}$$

Whenever a system undergoes a cyclic change, however complex the cycle may be, the algebraic sum of the work transfer is equal to the algebraic sum of the energy transfer as heat. Sign convention followed in this text:

- **Work done by a system** on its surroundings is treated as a **positive quantity**.
- **Energy transfer as heat to a system** from its surroundings is treated as a **positive quantity**

$$-\oint dQ - \oint(-dW) = \oint dQ - \oint dW = 0$$

1.19.1 Specific Heat at Constant Volume :

By definition it is the amount of energy required to change the temperature of a unit mass of the substance by one degree. $c_v = \left(\frac{dq}{dT}\right)_v$

While the volume is held constant. For a constant volume process, first law of thermodynamics gives

$dU = dQ \quad or, \quad du = dq$

Therefore, $c_v = \left(\frac{\partial u}{\partial T}\right)_v$

Where u is the specific internal energy of the system. If c_v varies with temperature, one can use mean specific heat at constant volume $\bar{c}_v = \left[\int_{T_1}^{T_2} c_v dT\right] / (T_2 - T_1)$

The total quantity of energy transferred during a constant volume process when the system temperature changes from T_1 to T_2, $Q = m\int_{T_1}^{T_2} c_v dT = U_2 - U_1$

The unit for c_v is kJ/kgK. The unit of molar specific heat is kJ/kmol K.

1.19.2. Specific Heat at Constant Pressure:

Specific heat at constant pressure is defined as the quantity of energy required to change the temperature of a unit mass of the substance by one degree during a constant pressure process.

$$c_p = \left(\frac{dq}{dT}\right)_p = \left(\frac{\partial h}{\partial T}\right)_p$$

The total heat interaction for a change in temperature from $T_1$to T_2 can be calculated from

$$Q = m\int_{T1}^{T2} c_p dT = H_2 - H_1$$

The molar specific heat at constant pressure can be defined as $\hat{c}_p = \left(\frac{\partial \hat{h}}{\partial T}\right)_p$

1.19.3. First law analysis of non-flow processes: The first law of thermodynamics can be applied to a system to evaluate the changes in its energy when it undergoes a change of state while interacting with its surroundings. The processes that are usually encountered in thermodynamic analysis of systems can be identified as any one or a combination of the following elementary processes:

- Constant volume (isochoric) process
- Constant pressure (isobaric) process
- Constant temperature (isothermal) process.
- Adiabatic process.
- Polytropic process.

The converse of the above statement is also true, i.e., there can be no machine which would continuously consume work without some other form of energy appearing simultaneously.

Process	PVT Relation	Work done	Internal energy	Heat transfer
Isobaric process	$\frac{V_1}{T_1}=\frac{V_2}{T_2}$	$P(V_2-V_1)=mR(T_2-T_1)$	$mC_v(T_2-T_1)$	$mC_p(T_2-T_1)$
Isochoric process	$\frac{P_1}{T_1}=\frac{P_2}{T_2}$	Zero	$mC_v(T_2-T_1)$	$mC_v(T_2-T_1)$
Isothermal process	$P_1V_1=P_2V_2$	$P_1V_1\ln\frac{V_2}{V_1}=P_1V_1\ln\frac{P_1}{P_2}=mRT_1\ln\frac{V_2}{V_1}$	Zero	$W_{1-2}=Q_{1-2}$
Adiabatic process	$\frac{T_2}{T_1}=\left(\frac{P_2}{P_1}\right)^{\gamma-1/\gamma}$ $\frac{T_2}{T_1}=\left(\frac{V_1}{V_2}\right)^{\gamma-1}$ $\left(\frac{P_1}{P_2}\right)=\left(\frac{V_2}{V_1}\right)^{\gamma}$	$W_{1-2}=\frac{P_1V_1-P_2V_2}{\gamma-1}=\frac{mR(T_1-T_2)}{\gamma-1}$	$mC_v(T_2-T_1)$	Zero

Polytropic Process	$\frac{T_2}{T_1}=\left(\frac{P_2}{P_1}\right)^{n-1/n}$ $\frac{T_2}{T_1}=\left(\frac{V_1}{V_2}\right)^{n-1}$ $\left(\frac{P_1}{P_2}\right)=\left(\frac{V_2}{V_1}\right)^{n}$	$W_{1-2}=\frac{P_1V_1-P_2V_2}{n-1}=\frac{mR(T_1-T_2)}{n-1}$	$mC_v(T_2-T_1)$	$Q_{1-2}=\frac{(\gamma-n)}{(\gamma-1)}*W_{\text{polytropic}}$

In the preceding sections we have developed first law of thermodynamics for a control mass going through a process. Many applications in thermodynamics are control volume. The present chapter is concerned with the development of control volume. Forms of the conservation of mass and energy in situations where there are flows are present.

SECOND LAW OF THERMODYNAMICS

1.21. Limitations of first law:

- First law fixes the exchange rate between heat & work, and places no restrictions on the direction of change.
- Process proceeds spontaneously in certain directions, but the reverse is not automatically attainable even though the reversal of the processes doesn't violet the first law.
- First law provides a necessary but not sufficient condition for a process to occur.
- There exists some directional law which would tell whether a particular process occurs or not.
- The first law fails to state the conditions under which energy conversions are possible. The law presumes that any change of thermodynamic state can take place in either direction, however, that is not true.

Ex: (i) Electric current flowing through a resistor produces heat electric current once dissipated as heat cannot be converted back to electricity.

(ii) Water flows from a higher level to a lower level & the reverse is not automatically possible. A mechanical energy from the external source would be required to pump the water back from the lower level to higher level.

1.21.1. Introduction to second law:

Second law states that, the process proceeds in a certain direction & not in the reverse direction. The use of the second law of thermodynamics is not limited to identifying the direction of processes. The second law also asserts that energy has quality as well as quantity. Second law provides necessary means to determine the quality as well as the degree of degradation of energy during a process.

Ex: - consider the heating of a room by the passage of electric current through a resistor. The first law dictates that the amount of electric energy supplied to the resistance wires be equal to the amount of energy transferred to the room as heat. But if we attempt the reverse process, it will come as no surprise that transferring some heat to the wires doesn't cause an equivalent amount of electric energy to be generated in the wires.

1.21.2. Thermal Reservoirs:

A thermal reservoir is defined as a large body of infinite heat capacity that can supply or absorb finite amount of heat without undergoing any change in temperature.

Ex: - large bodies of water such as oceans, lakes & rivers as well as the atmospheric air can be modeled accurately as thermal reservoir because of their large thermal energy storage capabilities or thermal masses.

1.21.3. Source:

The thermal energy reservoir from which heat is transferred to the system or the reservoir that supplies energy in form of heat s called source.

1.21.4. Sink:

The thermal energy reservoir to which heat is rejected from the system or the reservoir that absorbs energy in form of heat is called sink.

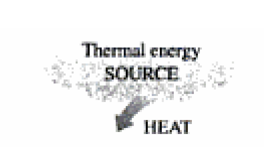


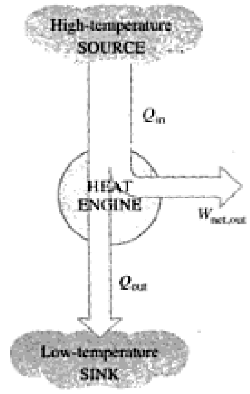
Fig .1.46. Source supplies energy in the form of heat & a sink absorbs it

1.21.5. Heat Engine: - A heat engine is a thermodynamic cycle in which there is net heat transfer to the system & a net work transfer from the system. The system which executes a heat engine cycle is called a heat engine.

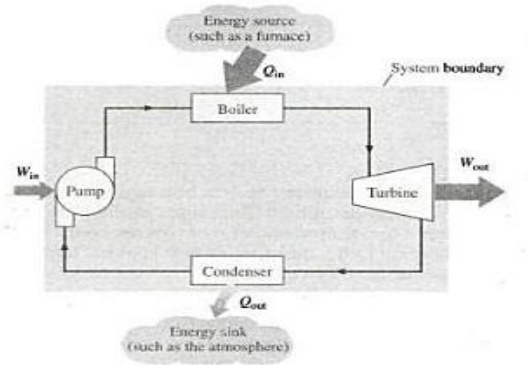
Characteristics of heat engine:

I. They receive heat from high temperature source (solar energy, oil furnace, nuclear reactor etc.)
II. They convert part of heat to work (usually inform of rotating shaft)
III. They reject the remaining waste heat to a low temperature sink.
IV. They operate on a cycle.

Heat engine & other cyclic derives usually involve a fluid to & from which heat is transferred while undergoing a cycle. The fluid is called the working fluid.



(a)



(b)

Fig.1.47 .Energy interaction in a heat engine

Q_1 = heat transferred to the system

Q_2 = heat rejected from the system

W_E= work done by the system

W_C = work done upon the system

W_T = turbine work

W_P = work done on the pump

In Fig 5. 2 (b)

Heat is transferred from the furnace to the water in the boiler to form steam which then works on the turbine rotor to produce work W_T, then the steam is condensed to water in the condenser in which on amount Q_2 is rejected from the system, & finally work W_p is done on the system (water) to pump it to the boiler. The system repeats the cycle.

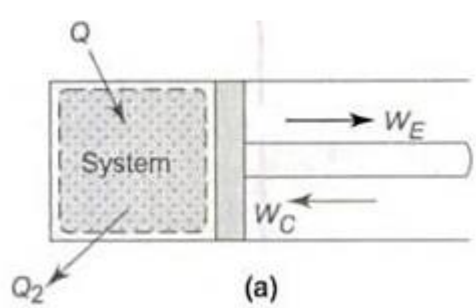


Fig 1.48 .Heat engine cycle performed by a closed system

In Fig.5.3, heat(Q_2) is transferred to the system, work W_E is done by the system, work W_C is done upon the system, & then heat Q_2 is rejected from the system. The system is brought back to initial state through all these four successive processes which constitute a heat engine cycle.

The net heat transfer in a cycle to either if the heat engines.

$Q_{net} = Q_1 - Q_2$------------------------------ (1)

& Net work transfer in a cycle

$W_{net} = W_T - W_P$----------------------------- (2)

(or $W_{net} = W_E - W_C$)

By the first law of thermodynamics, we have

$\Sigma_{cycle}Q = \Sigma_{cycle}W$

$Q_{net} = W_{net}$

$\Rightarrow Q_1 - Q_2 = W_T - W_P$-------------------------(3)

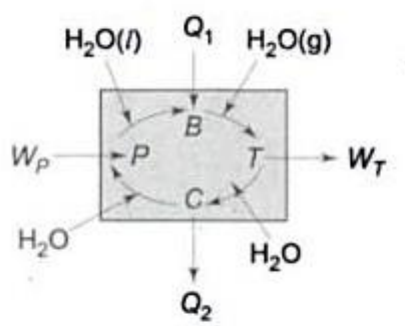


Fig .1.50. Cyclic heat engine with energy interaction s represented in a block diagram

Fig. 4 represents a cyclic heat engine in the form of block diagram indicating various energy interactions during a cycle. Boiler (B), Turbine (T), condenser (C) & Pump (P), all four together a heat engine.

Efficiency of a heat engine or heat engine cycle is defined as,

$$\eta = \frac{\text{Net work output of the cycle}}{\text{Total heat input to the cycle}}$$

$= \frac{Wnet}{Q1}$---------------------------- (4)

From equation 1,2,3 & 4,

$$\eta = \frac{\text{Wnet}}{\text{Q1}} = \frac{W_T - W_P}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

$$\Rightarrow \eta = 1 - \frac{Q_2}{Q_1}$$

This is also known as thermal efficiency of heat engine.

1.21.6. Kelvin- Plank statement of second law:

Kelvin-Plank statement of second law states, "It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature".

Efficiency of a heat engine is given by,

$$\eta = \frac{\text{Wnet}}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

A heat engine can never be 100% efficiency. Therefore, $Q_2>0$, i.e. there is always be a heat rejection. To produce net work in a thermodynamic cycle, a heat engine has thus to exchange with two reservoirs, the source & the sink.

1.21.7. PMM2:

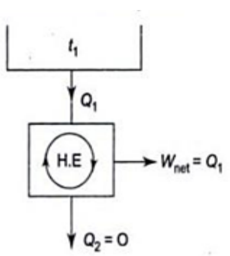


Fig 1.51 (A PMM2)

$$\eta = \frac{\text{Wnet}}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

If $Q_2= 0$ (i.e. $W_{net} = Q_1$ or $\eta = 1$), the heat engine will produce net work in a complete cycle by exchanging heat with only one reservoir, thus violating Kelvin-plank statement (Fig-5). Such a heat engine is called perpetual motion machine of second kind, (PMM2). A PMM2 is impossible.

1.21.8. Clausius' statement of the second law:

Clausius statement of second law states, "It is impossible to construct a device that operates in a cycle & produce no effect other than the transfer of heat from lower-temperature body to a higher temperature body".

1.21.9. Refrigerator:

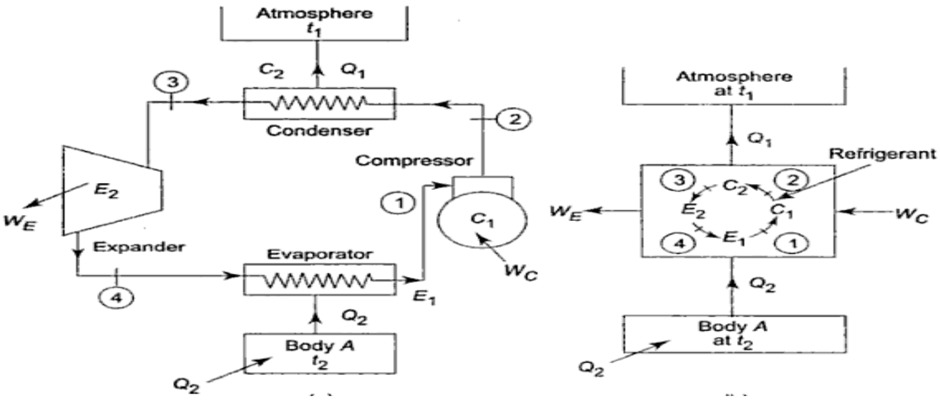


Fig 1.52 (a) **Fig 1.52 (b)**

A refrigerator is a device which operating in a cycle, maintains a body at a temperature lower than the temperature of surroundings.

Let t_1 = Ambient temperature

t_2 = temperature of body A.

Even though A is insulated, there always be heat leakage Q_2 into the body from surrounding by virtue of temperature difference.

Q_2 = designed out put

Q_1 = heat rejected from refrigeration to atmosphere.

The efficiency of refrigerator is expressed in terms of the co-efficient of performance (COP) . The objective of refrigerator is to remove heat (Q_2) from the refrigerated space.

The COP is refrigeration,

$$COP_R = \frac{\text{Desined output}}{\text{Required input}}$$

$$\Rightarrow COP_R = \frac{Q_2}{W_{net,in}} = \frac{Q_2}{Q_1 - Q_2} = \frac{1}{Q_1/Q_2 - 1}$$

1.21.10. Heat pump: Heat pump transfers heat from low temperature medium to high temperature medium. It is a device which operating in a cycle, maintains a body B, at a temperature higher than the temperature of the surroundings.

$COP_{HP} = \frac{\text{Desined output}}{\text{Required input}} = \frac{Q1}{Wnet.in}$

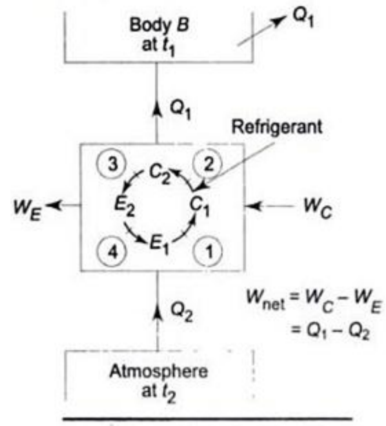


Fig. 1.53. A Cyclic heat pump

$\Rightarrow COP_{HP} = \frac{Q_1}{Q_1 - Q_2} = \frac{1}{1 - Q_2/Q_1}$ -------------------1

But $COP_R = \frac{Q_2}{Q_1 - Q_2}$ ------------------------2

Comparing equation 1 & 2, we get, $COP_{HP} = COP_R + 1$

1.21.11. Carnot Cycle: Carnot cycle is a reversible cycle, first proposed French Engineer Sadi Carnot in 1824. The theoretical heat engine that operates on Carnot cycle is called Carnot heat engine. The Carnot cycle is composed of four reversible process two isothermal & two adiabatic & it can be executed either in a closed or a steady flow system.

Consider a closed system that consists of a gas contained in an adiabatic piston-cylinder device, as shown in fig. The insulation of the cylinder head is such that it may be removed to bring cylinder into contact with reservoirs to provide heat transfer. The four reversible processes in Carnot cycle are,

(i) Reversible isothermal expansion: (Process 1-2, T_H= constant)

At state 1, the temperature of gas T_H & the cylinder head is in close contact with a source at temperature T_H. The gas is allowed to expand slowly, during work on surroundings. As the gas expands, the temperature of the gas tends to decrease. But as soon as the temperature drops by an infinitesimal amount dT, some heat is transferred from the reservoir into the gas, raising gas temperature to T_H. Thus, the gas temperature between the gas & he reservoir never exceeds a differential amount dT, this is a reversible heat transfer process. It continues until the piston reaches piston2. The amount of total heat transferred to the gas during this process is Q_1.

(ii) Reversible adiabatic expansion: - (Process 2-3, temperature drops from T_H to T_L). At state 2, the reservoir that was in contact with the cylinder head is removed & replaced by insulation so that the system becomes adiabatic. The gas continues to expand slowly, doing work on the surroundings until its temperature drops from T_H to T_2 (state 3). The piston is assumed to be frictionless & the process to be quasi-equilibrium, so the process is reversible as well as adiabatic.

(iii)Reversible Isothermal Compression: - (Process 3-4, T_L= Constant). At state 3, the insulation of the cylinder head is removed and the cylinder is brought into contact with sink temperature T_2. Now the piston is compressed, its temperature tends to rise. But as soon as it rises by an infinitesimal amount dT, heat is transferred from the gas to the sink, causing the gas temperature to drop T_L. Thus the gas temperature remains constant at T_L. Since the temperature differential between the gas and sink never exceeds a process. It continues until the piston reaches state 4. The amount of heat rejected from the gas during this process is Q_2.

(iv) Reversible adiabatic compression: - (Process 4-1, temperature rises from T_L to T_H). State 4 is such that when the low temperature reservoir is removed, the insulation put back on the cylinder head & the gas compressed in reverse manner, the gas returns to its initial state (stage 1) . The temperature rises from T_L to T_H during this reversible adiabatic compressor process, which completes the cycle.

P-V-diagram of the Carnot cycle:

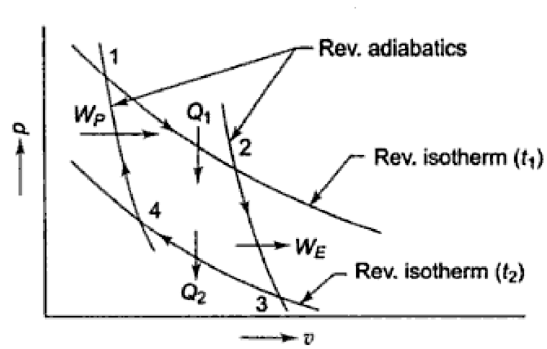


Fig.1.54 Carnot cycle on P-V diagram

Reversed heat engine:- Since all the process of the Carnot cycle are reversible it is possible to imagine that each process in the cycle is reversed & carried out in reversed order.

APPLICATION OF THERMODYNAMICS

1.23.1. STEAM POWER PLANT: Refer to Fig. 8.2. The layout of a modern steam power plant comprises of the following four circuits:

1. Coal and ash circuit.
2. Air and gas circuit.
3. Feed water and steam flow circuit.
4. Cooling water circuit.

Coal and Ash Circuit: Coal arrives at the storage yard and after necessary handling, passes on to the furnaces through the fuel feeding device. Ash resulting from combustion of coal collects at the back of the boiler and is removed to the ash storage yard through ash handling equipment.

Air and Gas Circuit: Air is taken in from atmosphere through the action of a forced or induced draught fan and passes on to the furnace through the air preheater, where it has been heated by the heat of flue gases which pass to the chimney via the preheater. The flue gases after passing around boiler tubes and super heater tubes in the furnace pass through a dust catching device or precipitator, then through the economizer, and finally through the air preheater before being exhausted to the atmosphere.

Feed Water and Steam Flow Circuit: In the water and steam circuit condensate leaving the condenser is first heated in a closed feed water heater through extracted steam from the lowest pressure extraction point of the turbine. It then passes through the deaerator and a few more water heaters before going into the boiler through economizer. In the boiler drum and tubes, water circulates due to the difference between the density of water in the lower temperature and the higher temperature sections of the boiler. Wet steam from the drum is further heated up in the super heater for being supplied to the prime mover. After expanding in high pressure turbine steam is taken to the reheat boiler and brought to its original dryness or superheat before being passed on to the low pressure turbine. From there it is exhausted through the condenser into the hot well. The condensate is heated in the feed heaters using the steam trapped (blow steam) from different points of turbine. A part of steam and water is lost while passing through different components and this is compensated by supplying additional feed water. This feed water should be purified before hand, to avoid the scaling of the tubes of the boiler.

Cooling Water Circuit: The cooling water supply to the condenser helps in maintaining a low pressure in it. The water may be taken from a natural source such as river, lake or sea or the same water may be cooled and circulated over again. In the latter case the cooling arrangement is made through spray pond or cooling tower.

1.23.2: Components of a Modern Steam Power Plant

A modern steam power plant comprises of the following components:

1. Boiler

(i) Super heater (ii) Re heater (iii) Economizer (iv) Air-heater.

2. Steam turbine 3. Generator 4. Condenser 5. Cooling towers 6. Circulating water pump 7. Boiler feed pump 8. Boiler chimney 9. Forced draught fans

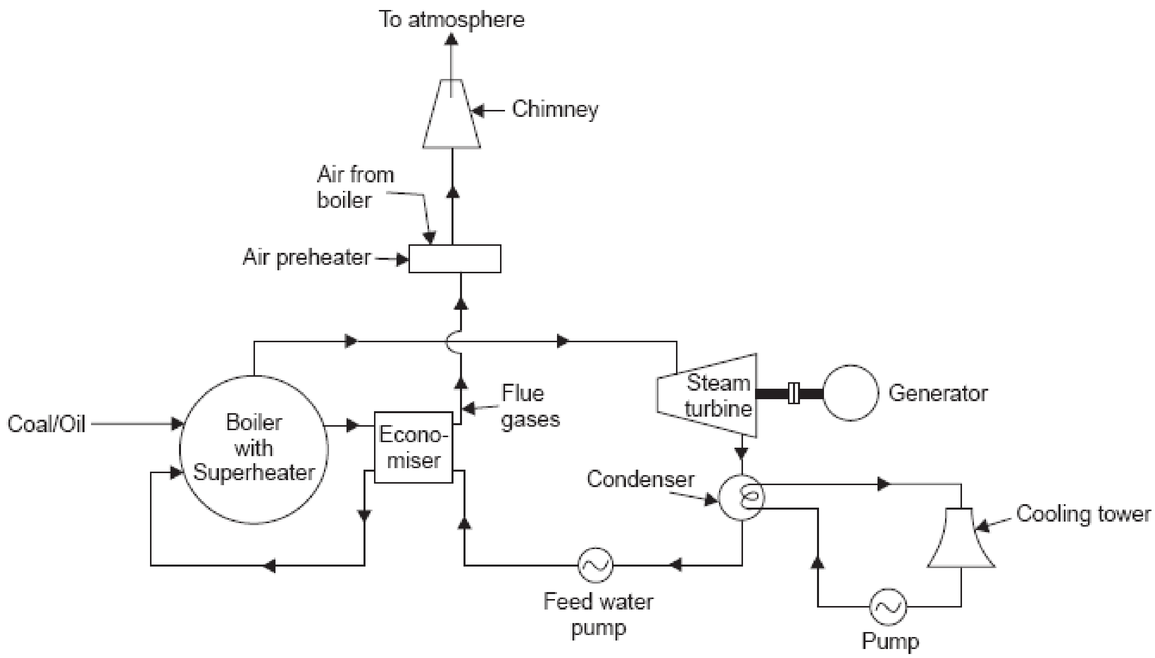


Fig.1.61.Layout of steam power plant

1.23.3. Description of Different parts of steam power plant:

Boiler: Water is converted into wet steam.

Turbine. Steam at high pressure expands in the turbine and drives the generator.

Condenser: It condenses steam used by the steam turbine. The condensed steam (known as ondensate) is used as feed water.

. **Cooling tower:** It cools the condenser circulating water. Condenser cooling water absorbs heat from eam. This heat is discharged to atmosphere in cooling water.

. **Condenser circulating water pump:** It circulates water through the condenser and the cooling tower.

. **Feed water pump:** It pumps water in the water tubes of boiler against boiler steam pressure.

To improve the efficiency of the power plant numerous auxiliary equipments and accessories are used

1.23.4. Refrigerator: Refrigeration uses the principle of **second law of thermodynamics**. According to the second law of thermodynamics; heat can flow from low temperature body to high temperature with some external work. Refrigerator is that mechanical device which keeps a body below atmospheric temperature. In order to operate the machine, some external work (electrical energy to the compressor of the refrigerator system is supplied). This helps the device to absorb heat from the body. The total heat absorbed along with work supplied is then rejected to the atmosphere in the form of heat.

Refrigeration: Refrigeration is defined as the process of continuous withdrawal of heat, from a body or within a space and to maintain it at a temperature lower than that which would exist under the natural influence of surroundings.

Refrigerator: Refrigerator is a mechanical device used for producing and maintaining the temperature of a body or space below ambient temperature.

Refrigerant: Refrigerant is the working fluid used in the refrigerator. Usually, the working fluid may be liquid or gas. The refrigerant should have a very low boiling point so that vaporizes on absorbing heat from the body to be cooled.

Types of Refrigeration system

1. Vapour compression Refrigeration system

2. Vapour absorption Refrigeration system

1.23.5. **Vapour compression Refrigeration system:** In a simple vapour compression system the following fundamental processes are completed in one cycle:

Expansion 2. Vaporization 3. Compression 4. Condensation

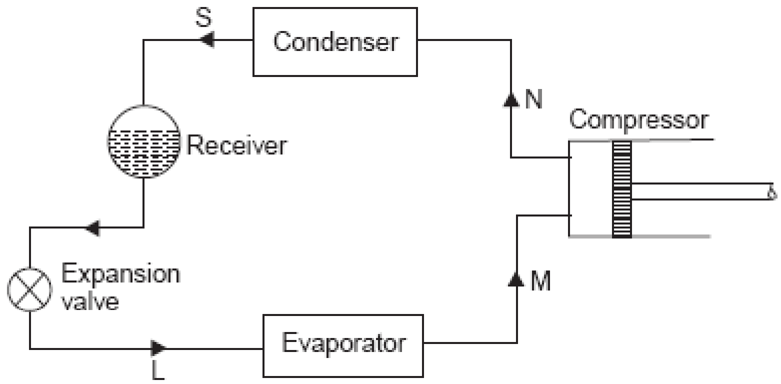


Fig.1.62. Simple vapour compression Refrigeration system.

The *vapour at low temperature and pressure* (state '*M*') enters the *compressor* where it is compressed isoentroprically and subsequently its temperature and pressure increase considerably (state '*N*'). This vapour after leaving the compressor enters the *condenser* where it is condensed into *high pressure liquid* (state '*S*') and is collected in a *receiver*. From receiver it passes through the *expansion valve*, here it is *throttled down to a lower pressure* and has *a low temperature* (state '*L*'). After finding its way through *expansion valve* it finally passes on to *evaporator* where it *extracts heat from the surroundings and vaporizes to low pressure vapour* (state '*M*').

1.23.6. Heat Pump: A **heat pump** is an electrical device that extracts heat from one place and transfers it to another. Refrigerators and air conditioners are both common examples of this technology. Heat pumps transfer heat by circulating a substance called a refrigerant through a cycle of evaporation and condensation (see graphic above). A compressor pumps the refrigerant between two heat exchanger coils. In one coil, the refrigerant is evaporated at low pressure and absorbs heat from its surroundings. The refrigerant is then compressed en route to the other coil, where it condenses at high pressure. At this point, it releases the heat it absorbed earlier in the cycle.

Refrigerators and air conditioners are both examples of heat pumps operating only in the cooling mode. A refrigerator is essentially an insulated box with a heat pump system connected to it. The evaporator coil is located inside the box, usually in the freezer compartment.

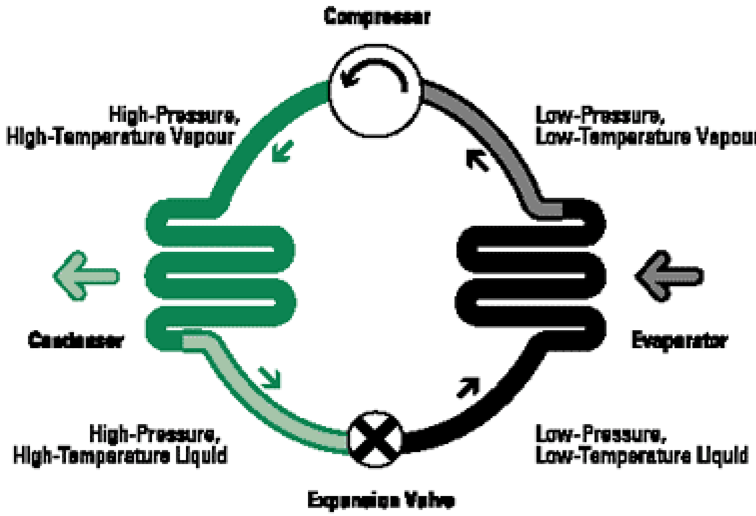


Fig.1.63. Heat Pump

Heat is absorbed from this location and transferred outside, usually behind or underneath the unit where the condenser coil is located. Similarly, an air conditioner transfers heat from inside a house to the outdoors. The heat pump cycle is fully reversible, and heat pumps can provide year-round climate control for your home – heating in winter and cooling and dehumidifying in summer. Since the ground and air outside always contain some heat, a heat pump can supply heat to a house even on cold winter days. In fact, air at –18°C contains about 85 percent of the heat it contained at 21°C.